Test Series No.

1.4

MASTERY TEST SERIES

DURATION: 90 Minutes | **M. MARKS: 400**

DATE: 13/02/2026

ANSWER KEY

1. (1)	26. (4)	51. (1)	76. (3)
2. (3)	27. (4)	52. (3)	77. (2)
3. (2)	28. (4)	53. (2)	78. (2)
4. (2)	29. (1)	54. (2)	79. (1)
5. (1)	30. (3)	55. (2)	80. (1)
6. (4)	31. (4)	56. (3)	81. (3)
7. (4)	32. (1)	57. (1)	82. (1)
8. (2)	33. (1)	58. (2)	83. (3)
9. (2)	34. (3)	59. (2)	84. (3)
10. (3)	35. (3)	60. (3)	85. (1)
11. (3)	36. (3)	61. (2)	86. (2)
12. (1)	37. (1)	62. (1)	87. (2)
13. (2)	38. (3)	63. (4)	88. (1)
14. (3)	39. (3)	64. (2)	89. (2)
15. (2)	40. (3)	65. (4)	90. (1)
16. (1)	41. (3)	66. (2)	91. (4)
17. (4)	42. (2)	67. (4)	92. (2)
18. (2)	43. (1)	68. (1)	93. (3)
19. (3)	44. (3)	69. (4)	94. (4)
20. (2)	45. (4)	70. (3)	95. (4)
21. (4)	46. (2)	71. (1)	96. (3)
22. (4)	47. (1)	72. (3)	97. (2)
23. (2)	48. (4)	73. (3)	98. (2)
24. (3)	49. (1)	74. (2)	99. (1)
25. (2)	50. (4)	75. (3)	100. (2)

[Contd...

1. (1)

The density of electric field lines represents the magnitude of the electric field strength. In regions where the field lines are closer together (more dense), the electric field is stronger.

(NCERT Ed. 2025-26, 12th, Page No. 19)

2. (2)

$$\eta = \frac{F \cdot \Delta z}{A \cdot \Delta v}$$

$$[\eta] = \frac{[MLT^{-2}] \cdot [L]}{[L^2] \cdot [LT^{-1}]}$$

$$[\eta] = \frac{[ML^2T^{-2}]}{[L^3T^{-1}]}$$

$$[\eta] = [M] \cdot [L^{2-3}] \cdot [T^{-2-(-1)}]$$

$$[\eta] = [ML^{-1}T^{-1}]$$

(NCERT Ed. 2025-26, 11th, Page No. 191)

3. (2)

- Height of Ball 1: $h_1 = ut - \frac{1}{2}gt^2$
- Height of Ball 2: $h_2 = u(t-2) - \frac{1}{2}g(t-2)^2$

At the point of collision, their heights are equal ($h_1 = h_2$):

$$ut - \frac{1}{2}gt^2 = u(t-2) - \frac{1}{2}g(t-2)^2$$

Expand the right side:

$$ut - \frac{1}{2}gt^2 = ut - 2u - \frac{1}{2}g(t^2 - 4t + 4)$$

$$ut - \frac{1}{2}gt^2 = ut - 2u - \frac{1}{2}gt^2 + 2gt - 2g$$

Now, cancel the common terms (ut and $-\frac{1}{2}gt^2$) from both sides:

$0 = -2u + 2gt - 2g$

Rearrange to solve for t:

$2gt = 2u + 2g$

$gt = u + g$

$$t = \frac{u+g}{g} \Rightarrow t = \frac{u}{g} + 1$$

$$t = \frac{40}{10} + 1 = 4 + 1 = 5 \text{ seconds}$$

$$h = ut - \frac{1}{2}gt^2$$

$$h = (40 \times 5) - \frac{1}{2}(10 \times 5^2)$$

$h = 75$ meters

(NCERT Ed. 2025-26, 11th, Page No. 18)

4. (2)

In a uniform electric field, the field components are given by the negative gradient of the potential:

$$E_x = -\frac{\Delta V}{\Delta x} \text{ and } E_y = -\frac{\Delta V}{\Delta y}$$

For the x-component (E_x):

Using points A and D:

$\Delta V = V_D - V_A = 5V - 2V = 3V$

$\Delta x = 0.2$ m

$$E_x = -\frac{3V}{0.2m} = -15 \text{ V/m}$$

For the y-component (E_y):

Using points A and B:

$\Delta V = V_B - V_A = 5V - 2V = 3V$

$\Delta y = 0.2$ m

$$E_y = -\frac{3V}{0.2m} = -15 \text{ V/m}$$

$$E = \sqrt{E_x^2 + E_y^2}$$

$$E = \sqrt{(-15)^2 + (-15)^2} = \sqrt{225+225} = \sqrt{450}$$

$E = 15\sqrt{2}$ V/m

Direction:

Electric field lines point from higher potential to lower potential.

Looking at the values: V(C) = 8V, V(P) = 5V, and V(A) = 2V.

The potential decreases along the line from C through P to A. Therefore, the electric field vector points along the direction PA.

(NCERT Ed. 2025-26, 12th, Page No. 55)

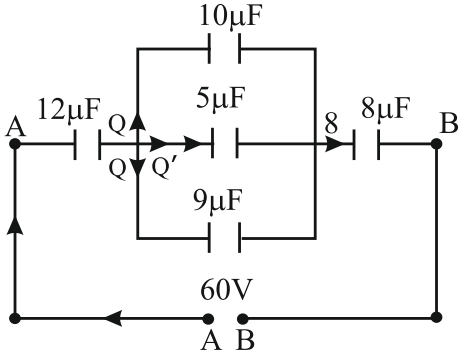
5. (1)

Current density is inversely proportional to the cross-sectional area of wire.

(NCERT Ed. 2025-26, 12th, Page No. 84)

6. (4)

The given circuit can be redrawn as follows:



[Contd...

Equivalent capacitance of the circuit, $C_{AB} = 4\mu F$
Charge given by the battery
$Q = C_{eq}V = 4 \times 60 = 240\mu C$
Charge in 5μF capacitor

$$Q' = \frac{5}{(10+5+9)} \times 240 = 50\mu C$$

(NCERT Ed. 2025-26, 12th, Page No. 72)

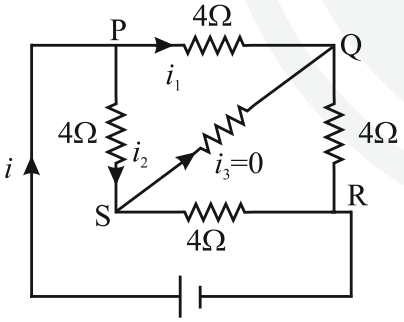
7. (4)

$F_m = q((3\hat{i}) \times (5\hat{j})) = 15q\hat{k}$
$F_e = qE = -15q\hat{k}$
$F_{net} = 15q\hat{k} - 15q\hat{k} = 0$
Net force on particle is zero, so it moves with constant velocity.

(NCERT Ed. 2025-26, 12th, Page No. 109)

8. (2)

Resistance of each side = 8 cm × 0.5 ohm/cm = 4 ohm.
Circuit Analysis: The battery is connected across diagonals P and R. Due to the symmetry of the square (all sides are 4 ohm), the circuit forms a Balanced Wheatstone Bridge.
In a balanced bridge, no current flows through the central diagonal QS. We can remove it.
Top branch (P-Q-R): 4 ohm + 4 ohm = 8 ohm.
Bottom branch (P-S-R): 4 ohm + 4 ohm = 8 ohm.
Equivalent Resistance (Req): Two 8 ohm branches in parallel result in 4 ohm.



Current $i = \frac{4}{4} = 1$ A

(NCERT Ed. 2025-26, 12th, Page No. 97)

9. (2)

Force per unit length between two wires carrying currents i_1 and i_2 at distance r is given by

$$\frac{F}{l} = \frac{\mu_o}{2\pi}\frac{i_1 i_2}{r}$$

Here, $i_1 = i_2 = i$
and $r = b$

$$\therefore \frac{F}{l} = \frac{\mu_o i^2}{2\pi b}$$

(NCERT Ed. 2025-26, 12th, Page No. 123)

10. (3)

$H_{total} = H + 2H = 3H$
Let a small hole be made at a height x from the ground. The depth of the hole from the top surface of the liquid is:
$h = 3H - x$
According to Torricelli's law, the velocity of the liquid coming out of the hole is:

$$v = \sqrt{2gh} = \sqrt{2g(3H-x)}$$

$$x = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2x}{g}}$$

$R = v \times t$

$$R = \sqrt{2g(3H-x)} \times \sqrt{\frac{2x}{g}}$$

$$R = \sqrt{4x(3H-x)} = 2\sqrt{3Hx - x^2}$$

$$\frac{d}{dx}(3Hx - x^2) = 0$$

$3H - 2x = 0$
$2x = 3H$

$$x = \frac{3H}{2} = 1.5H$$

(NCERT Ed. 2025-26, 11th, Page No. 189)

11. (3)

$W = KE_{final} - KE_{initial}$
Since the particle starts from rest, the initial kinetic energy ($KE_{initial}$) is 0. Therefore:
Kinetic Energy at 3 m = Work done from 0 to 3 m
Part 1: Rectangle (from 0 to 2 m)
Base = 2 m
Height = 2 N
$Area_1$ = Base × Height = 2 × 2 = 4 J
Part 2: Trapezium (from 2 to 3 m)
Width = 3 – 2 = 1 m
Parallel side 1 (at 2m) = 2 N
Parallel side 2 (at 3m) = 3 N

$$Area_2 = \frac{1}{2} \times (\text{sum of parallel sides}) \times \text{width}$$

$$Area_2 = \frac{1}{2} \times (2+3) \times 1 = \frac{1}{2} \times 5 = 2.5 \text{ J}$$

Total Work (W) = $Area_1 + Area_2$
W = 4 J + 2.5 J = 6.5 J

(NCERT Ed. 2025-26, 11th, Page No. 76)

12. (1)

Using KVL
$5 - 14i - 4i + 15 - 2i = 0$, $i = 1$A
$V_{(across\ 14\Omega)} = 1 \times 14 = 14$ volt
$V_{(across\ cell\ A)} = 5 - 2i = 3$V
$V_{(across\ cell\ B)} = 15 - 4i = 11$V

(NCERT Ed. 2025-26, 12th, Page No. 95)

[Contd...

13. (2)

From Kepler's third law $T^2 \propto r^3$ or $T \propto (r)^{3/2}$

$$\therefore \frac{T_2}{T_1} = \left(\frac{r_2}{r_1}\right)^{3/2} \text{ or}$$

$$T_2 = T_1\left(\frac{r_2}{r_1}\right)^{3/2}$$

$$= (365)\left(\frac{1}{2}\right)^{3/2}$$

$\Rightarrow T_2 \approx 129$ days

(NCERT Ed. 2025-26, 11th, Page No. 137)

14. (3)

Thermal stress $\sigma = Y\ \alpha\Delta\theta$

Given,

$\sigma_1 = \sigma_2$

$\therefore\ Y_1 \alpha_1 \Delta\theta = Y_2 \alpha_2 \Delta\theta$

or $\frac{Y_1}{Y_2} = \frac{\alpha_2}{\alpha_1} = \frac{3}{2}$

(NCERT Ed. 2025-26, 11th, Page No. 207)

15. (2)

$$y_1 = 3\left[\sin 2\pi t + \sqrt{3}\cos 2\pi t\right]$$

$$= 6[\frac{1}{2}\sin 2\pi t + \frac{\sqrt{3}}{2}\cos 2\pi t\ ]$$

$$= 6[\cos\frac{\pi}{3}\sin 2\pi t + \sin\frac{\pi}{3}\cos 2\pi t]$$

$$= 6\sin(2\pi t + \frac{\pi}{3}) \Rightarrow A_1 = 6$$

Similarly,

$$y_2 = 3\sin\left(2\pi t + \frac{\pi}{4}\right)$$

$\Rightarrow A_2 = 3$

Hence, $\frac{A_1}{A_2} = \frac{6}{3} = \frac{2}{1}$

(NCERT Ed. 2025-26, 11th, Page No. 262)

16. (1)

Heat required

$Q_1 = (1.1 + 0.02) \times 10^3 \times 1 \times (80 - 15)$

$= 72800$ cal

Heat given it m (in kg) steam is condensed:

$Q_2 = (m \times 540 \times 10^3) + m \times 1 \times 10^3 \times (100 - 80)$

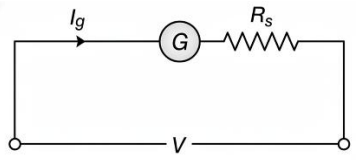
Equating $Q_1 = Q_2$, we get

$m = 0.130$ kg

(NCERT Ed. 2025-26, 11th, Page No. 213)

17. (4)

For a voltmeter,



$I_g(G + R_s) = V \Rightarrow R_s = \frac{V}{I_g} - G$

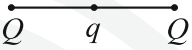
$\Rightarrow\ R_s = 1985 = 1.985\ k\Omega$

or $R = 1.985 \times 10^3\ \Omega$

(NCERT Ed. 2025-26, 12th, Page No. 131)

18. (2)

Since, q is at the centre of two charges Q and Q, net force on it is zero, whatever the magnitude and sign of charge



on it. For the equilibrium of Q, q should be negative because other charge Q will repel it, so q should attract it. At the same time, these attractions and repulsions should be equal.

$$\frac{1}{4\pi\epsilon_o}\frac{QQ}{r^2} = \frac{1}{4\pi\epsilon_o}\frac{Qq}{(r/2)^2}$$

- or $q = \frac{Q}{4}$ or with sign $q = -\frac{Q}{4}$

(NCERT Ed. 2025-26, 12th, Page No. 13)

19. (3)

ΔU = decrease in potential energy

$= U_i - U_f$

$$= \frac{1}{2}C(V_1^2 + V_2^2) - \frac{1}{2}(2C)\left(\frac{V_1 + V_2}{2}\right)^2$$

$$= \frac{1}{4}C(V_1 - V_2)^2$$

(NCERT Ed. 2025-26, 12th, Page No. 76)

20. (2)

Since the two fuse wires have equal resistance (R) and are connected in parallel, the total current (I) entering the combination will divide equally between them.

$I_1 = I_2 = \frac{I}{2}$

Fuse 1 (Rated 10 A): It will blow if the current passing through it exceeds 10 A.

$\frac{I}{2} > 10 \Rightarrow I > 20A$

Fuse 2 (Rated 5 A): It will blow if the current passing through it exceeds 5 A.

[Contd...

$\frac{I}{2} > 5 \Rightarrow I > 10A$

The combination is safe only as long as neither fuse blows.

If we pass a total current of 10 A, both fuses carry 5 A. Fuse 2 is at its limit, and Fuse 1 is well within its limit.

If we try to pass more than 10 A (e.g., 12 A), each fuse receives 6 A. Fuse 2 (rated 5 A) will blow immediately. Once Fuse 2 blows, the entire 12 A will shift to Fuse 1, causing Fuse 1 (rated 10 A) to blow as well.

Therefore, the maximum safe current (effective rating) for the combination is governed by the weaker fuse.

Effective Rating = 10 A

(NCERT Ed. 2025-26, 12th, Page No. 93)

21. (4)

$T = \frac{2\pi m}{qB}$

Time period is independent of speed of the particle.

(NCERT Ed. 2025-26, 12th, Page No. 112)

22. (4)

Net charge enclosed inside spherical surface

$= +q - q = 0$

(NCERT Ed. 2025-26, 12th, Page No. 30)

23. (2)

Decrease in potential energy of rod = Gain of rotational kinetic energy

$\Rightarrow \; Mg \times \frac{L}{2} \sin 30°$

$= \frac{1}{2} \times I \times \omega^2$

$\Rightarrow \; Mg \times \frac{L}{2} \times \frac{1}{2} = \frac{1}{2} \times I \times \omega^2$

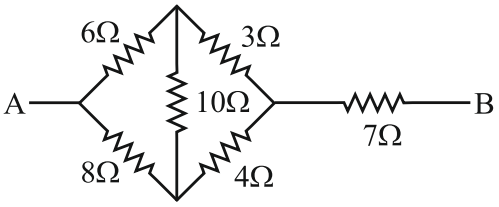
Here, $I = \frac{ML^2}{3}$

Substituting the values, we get

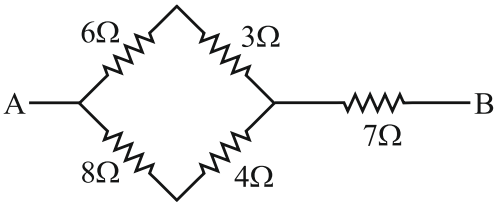
$\omega = \sqrt{30}$ rad/s

(NCERT Ed. 2025-26, 11th, Page No. 120)

24. (3)

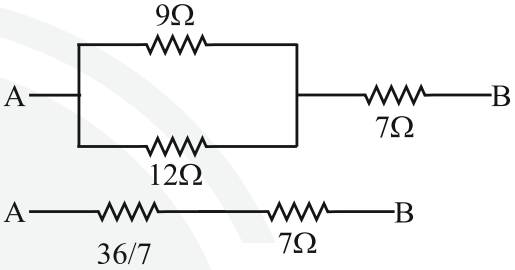


Wheat stone bridge.



Since 6Ω and 3Ω are in series

Since 86 Ω and 4Ω are in series



$R_{eq} = \frac{85}{7}\Omega$

(NCERT Ed. 2025-26, 12th, Page No. 99)

25. (2)

Since plates 1 and 4 are connected, they act as a single outer boundary.

Distance from Plate 2 to the left boundary (Plate 1) = d

Distance from Plate 2 to the right boundary (Plate 4) = 2d + d = 3d (Since Plate 3 is neutral, the total gap is just the sum of distances).

Charge Distribution Rule:

Charge on a plate divides between its two sides inversely proportional to the distances to the nearest grounded/connected plates.

Charge moving towards the right (Q right)

$= Q \times \frac{\text{Distance Left}}{\text{Total Distance}}$

$Q \text{ right} = Q \times \frac{d}{d+3d} = Q \times \frac{d}{4d} = \frac{Q}{4}$

Induction on Plate 3:

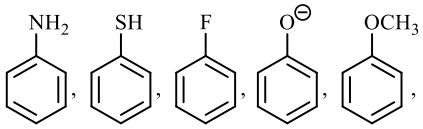
The charge +Q/4 on the right side of Plate 2 induces -Q/4 on the left side of Plate 3.

Since Plate 3 is neutral (Total charge = 0), an equal and opposite charge of +Q/4 must appear on its right side.

(NCERT Ed. 2025-26, 12th, Page No. 69)

[Contd...

26. (4)



(NCERT Ed. 2025-26 11th Part-II Page No. 278)

27. (4)

$MgCl_2 \rightleftharpoons Mg^{2+} + 2Cl^-,\ n = 3$

$i = 1 + (n-1)\alpha$; $2.8 = 1 + (3-1)\alpha$

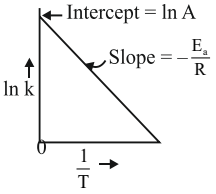
$\alpha = 0.9$; $\%\,\alpha = 0.9 \times 100 = 90\%$

(NCERT Ed. 2025-26 12th Part-I Page No. 24)

28. (4)

$k = A\,e^{-Ea/RT}$

$\ln k = -\frac{E_a}{RT} + \ln A$



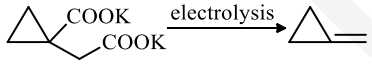
(NCERT Ed. 2025-26 12th Part-I Page No. 79)

29. (1)

- Lesser the steric hinderance on halide carbon, more will be reactivity towards S_N2 reaction.
- Reactivity of RX for S_N2 reaction:
  $1° > 2° > 3°$

(NCERT Ed. 2025-26 12th Part-II Page No. 174)

30. (3)



(NCERT Ed. 2025-26 11th Part-II Page No. 301)

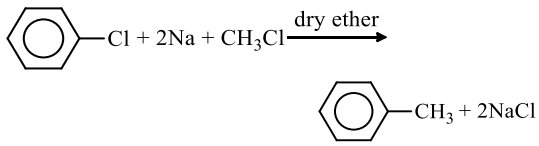
31. (4)

- In electrochemical cell, anode is negative terminal while cathode is positive terminal.
- The potential of standard hydrogen electrode is zero.

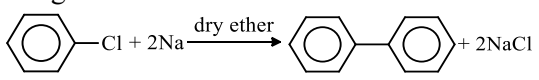
(NCERT Ed. 2025-26 12th Part-I Page No. 34)

32. (1)

- Wurtz Fittig reaction:



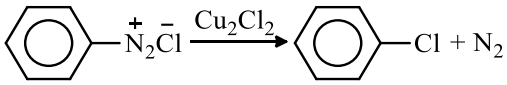
- Fittig reaction:



- Finkelstein reaction:

$CH_3CH_2Cl + NaI \rightarrow CH_3CH_2I + NaCl$

- Sandmeyer's reaction:



(NCERT Ed. 2025-26 12th Part-II Page No. 186)

33. (1)

$\Delta T_f = iK_f m = 2 \times 1.86 \times 1 = 3.72°C$

$T_f = T_f^0 - \Delta T_f = 0°C - 3.72°C = -3.72°C$

(NCERT Ed. 2025-26 12th Part-I Page No. 18)

34. (3)

Paper chromatography → stationary phase = water trapped in the paper

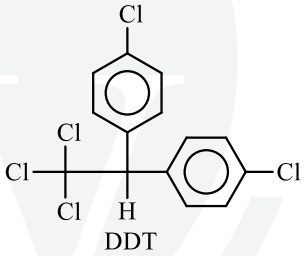
(NCERT Ed. 2025-26 11th Part-II Page No. 284)

35. (3)

Λ_m of CH_3COOH changes with dilution because on dilution degree of dissociation of CH_3COOH increases.

(NCERT Ed. 2025-26 12th Part-I Page No. 49)

36. (3)



(NCERT Ed. 2025-26 12th Part-II Page No. 188)

37. (1)

$t = \frac{2.303}{k} \log \frac{a}{a-x}$

$t = \frac{2.303}{2 \times 10^{-2}} \log \frac{100}{25}$

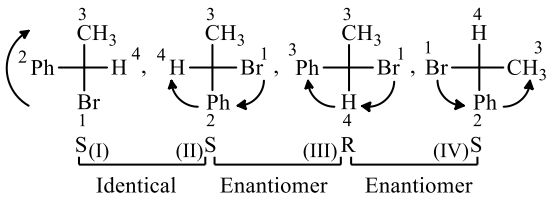
$t = \frac{2.303 \times 2 \times \log 2}{2 \times 10^{-2}}$

$t = 2.303 \times 0.301 \times 10^2$

$t = 69.3\ \text{sec}$

(NCERT Ed. 2025-26 12th Part-I Page No. 72)

38. (3)



(NCERT Ed. 2025-26 11th Part-II Page No. 272)

[Contd...

39. (3)

$p_t = p_A^0 + \left(p_B^0 - p_A^0\right)\chi_B$

$600 = 450 + (700 - 450)\chi_B; \quad \chi_B = 0.6$

So, $\chi_A = 1 - 0.6 = 0.4$

(NCERT Ed. 2025-26 12th Part-I Page No. 15)

40. (3)

$$E_{cell} = E^o_{cell} - \frac{0.059}{2}\log\left(\frac{[Mg^{2+}]}{[Ag^+]^2}\right)$$

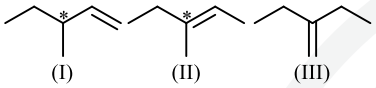
$$= 3.17 - \frac{0.059}{2}\log\frac{(10^{-3})}{(10^{-2})^2} = 3.14 \text{ V}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 38)

41. (3)

From molar mass, it can be C_6H_{12}

Molar mass = (6 × 12 + 12 × 1 = 84)



(I) (II) (III)

(I) Has chiral carbon and (II) show cis/trans

(NCERT Ed. 2025-26 11th Part-II Page No. 284)

42. (2)

When the concentration of Q is doubled, the rate is doubled. Hence, the order of reaction with respect to Q is one.

Further, when concentration of P is doubled, the rate remains unaltered. So, order of reaction with respect to P is zero.

rate = k[Q]

Overall order = 0 + 1 = 1

(NCERT Ed. 2025-26 12th Part-I Page No. 08)

43. (1)

- S_N2 reactions are stereospecific, indicating that they result in the formation of only one stereo-isomers as the product.
- S_N1 reactions involve formation of carbocations.

(NCERT Ed. 2025-26 12th Part-II Page No. 177)

44. (3)

In (3) $\overset{-}{C}H_2-\overset{+}{C}H-\overset{+}{N}\overset{O}{\overset{\|}{\diagup}}\underset{\diagdown}{O^-}$ two positive charge on adjacent posion.

(NCERT Ed. 2025-26 11th Part-II Page No. 00)

45. (4)

49% H_2SO_4 mass/volume means 49 g of H_2SO_4 is dissolved in 100 mL of solution.

Weight of solution = density × volume

= 1.49 × 100 = 149 g

Weight of solvent = 149 – 49 = 100 g

Molality = $= \frac{49}{98 \times 100} \times 1000 = 5 \text{ m}$

(NCERT Ed. 2025-26 12th Part-I Page No. 03)

46. (2)

$\Delta T_b \propto (i \times m)$

- 0.1 m Na_2SO_4 : 0.1 × 3 = 0.3
- 0.01 m $(NH_4)_3PO_4$: 0.01 × 4 = 0.04
- 0.5 m $BaCl_2$: 0.5 × 3 = 1.5
- 0.2 m $Al_2(SO_4)_3$: 0.2 × 5 = 1.0

(NCERT Ed. 2025-26 12th Part-I Page No. 22)

47. (1)

- Order is applicable to elementary as well as complex reactions whereas molecularity is applicable only for elementary reactions.
- For complex reaction molecularity has no meaning.

(NCERT Ed. 2025-26 12th Part-I Page No. 70)

48. (4)

$$n_A = \frac{180}{18} = 10$$

$$n_B = \frac{18}{180} = 0.1$$

$$\frac{p^0 - p_s}{p^0} = \frac{n_B}{n_A + n_B} = \frac{0.1}{0.1 + 10} = \frac{1}{101}$$

$$p^0 - p_s = \frac{p^0}{101}$$

$$p_s = p^0 - \frac{p^0}{101} = \frac{100\,p^0}{101}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 15)

49. (1)

Degree of dissociation,

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0} \Rightarrow \alpha = \frac{8.0}{400} = 2 \times 10^{-2}$$

From Ostwald's dilution law (for weak monobasic acid),

$$K_c = \frac{C\alpha^2}{(1-\alpha)} \quad (\because 1 >>> \alpha)$$

$$= \frac{1}{32}(2 \times 10^{-2})^2 = 1.25 \times 10^{-5}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 49)

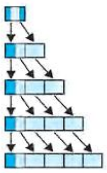
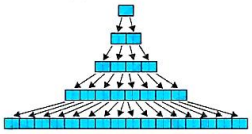
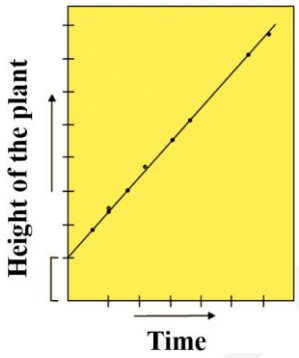
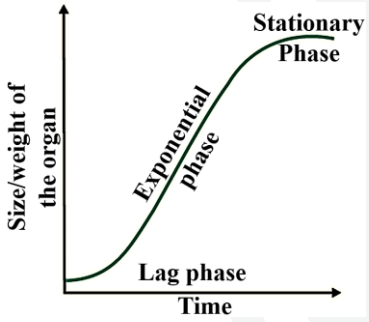
50. (4)

The reaction is electrophilic addition, because the first attacking species is the electrophile (H^+).

(NCERT Ed. 2025-26 11th Part-II Page No. 311)

[Contd...

51. (1)

	Arithmetic growth
	Geometric Growth
	Constant linear growth
	An idealised sigmoid growth curve

(NCERT Ed. 2025-26, 11th, Page No. 170)

52. (3)
In marginal placentation the placenta forms a ridge along the ventral suture of the ovary and the ovules are borne on this ridge forming two rows, as in pea. When the placenta is axial and the ovules are attached to it in a multilocular ovary, the placentation is said to be axile, as in china rose, tomato and lemon. In parietal placentation, the ovules develop on the inner wall of the ovary or on peripheral part. Ovary is one-chambered but it becomes two chambered due to the formation of the false septum, e.g., mustard and *Argemone*. When the ovules are borne on central axis and septa are absent, as in *Dianthus* and Primrose the placentation is called free central.

(NCERT Ed. 2022-23, 11th, Page No. 65)

53. (2)
The exponential growth can be expressed as $W_1 = W_0 e^{rt}$.
W_1 = final size (weight, height, number etc.), W_0 = initial size at the beginning of the period, r = growth rate, t = time of growth, e = base of natural logarithms

(NCERT Ed. 2025-26, 11th, Page No. 170)

54. (2)
In basidiomycetes, the asexual spores are generally not found, but vegetative reproduction by fragmentation is common. In ascomycetes, sexual spores are called ascospores which are produced endogenously in sac like asci.

(NCERT Ed. 2025-26, 11th, Page No. 18)

55. (2)
In gymnosperms the microspores develop into a male gametophytic generation which is highly reduced and is confined to only a limited number of cells. This reduced gametophyte is called a pollen grain.

(NCERT Ed. 2025-26, 11th, Page No. 33)

56. (3)
When the pollen grain is mature it contains two cells, the vegetative cell and generative cell. In over 60 per cent of angiosperms, pollen grains are shed at this 2-celled stage. In the remaining species, the generative cell divides mitotically to give rise to the two male gametes before pollen grains are shed (3-celled stage).

(NCERT Ed. 2025-26, 12th, Page No. 18)

57. (1)
2, 4-D, widely used to kill dicotyledonous weeds.

(NCERT Ed. 2025-26, 11th, Page No. 176)

58. (2)
In dicot stem the innermost layer of the cortex is called the endodermis. The cells of the endodermis are rich in starch grains and the layer is also referred to as the starch sheath. The monocot stem has a sclerenchymatous hypodermis. The phloem parenchyma is absent, and water-containing cavities are present within the vascular bundles

(NCERT Ed. 2022-23, 11th, Page No. 75)

59. (2)
Measurement and the comparison of total growth per unit time is called the absolute growth rate.
The growth of the given system per unit time expressed on a common basis, e.g., per unit initial parameter is called the relative growth rate.

(NCERT Ed. 2025-26, 11th, Page No. 171)

60. (3)
In the angiosperm embryo sac, six of the eight nuclei are surrounded by cell walls and organised into cells; the remaining two nuclei, called polar nuclei are situated below the egg apparatus in the large central cell.

(NCERT Ed. 2025-26, 12th, Page No. 20)

[Contd...

61. (2)
Ubiquinone receives reducing equivalents via $FADH_2$ (succinate dehydrogenase-complex II) that is generated during oxidation of succinate in the citric acid cycle. The reduced ubiquinone (ubiquinol) is then oxidised with the transfer of electrons to cytochrome *c* via cytochrome bc_1 complex (complex III). Cytochrome *c* is a small protein attached to the outer surface of the inner membrane and acts as a mobile carrier for transfer of electrons between complex III and IV. Complex IV refers to cytochrome *c* oxidase complex containing cytochromes *a* and a_3, and two copper centres.
(NCERT Ed. 2025-26, 11th, Page No. 160)

62. (1)
Ethylene initiates germination in peanut seeds. ABA inhibits seed germination. Cytokinins help overcome the apical dominance. Spraying juvenile conifers with GAs hastens the maturity period, thus leading to early seed production. make match the pair
(NCERT Ed. 2025-26, 11th, Page No. 177)

63. (4)
Some plants such as *Viola* (common pansy), *Oxalis*, and *Commelina* produce two types of flowers – chasmogamous flowers which are similar to flowers of other species with exposed anthers and stigma, and cleistogamous flowers which do not open at all.
(NCERT Ed. 2025-26, 12th, Page No. 22)

64. (2)
Class, this category includes related orders. In case of animals, classes with a few similar characters are assigned to a higher category called phylum. In case of plants, classes with a few similar characters are assigned to a higher category called Division. All animals belonging to various phyla are assigned to the highest category called Kingdom Animalia in the classification system of animals. The Kingdom Plantae, on the other hand, is distinct, and comprises all plants from various divisions.
(NCERT Ed. 2022-23, 11th, Page No. 7, 8)

65. (4)
Mitosis usually results in the production of diploid daughter cells with identical genetic complement.
Mitotic divisions in the meristematic tissues – the apical and the lateral cambium, result in a continuous growth of plants throughout their life.
(NCERT Ed. 2025-26, 11th, Page No. 125)

66. (2)
Apomixis is a form of asexual reproduction that mimics sexual reproduction.
(NCERT Ed. 2025-26, 12th, Page No. 28)

67. (4)
In human beings, the membrane of the erythrocyte has approximately 52 per cent protein and 40 per cent lipids.
(NCERT Ed. 2025-26, 11th, Page No. 93)

68. (1)
Embryos of monocotyledons possess only one cotyledon. In the grass family the cotyledon is called scutellum. The portion of the embryonal axis above the level of attachment of scutellum is the epicotyl. Epicotyl has a shoot apex and a few leaf primordia.
(NCERT Ed. 2022-23, 11th, Page No. 67)

69. (4)
The fixation of six molecules of CO_2 and 6 turns of the Calvin cycle are required for the formation of one molecule of glucose from the pathway. For the formation of three glucose molecules 18 turns(6 X 3) of the Calvin cycle are required.
(NCERT Ed. 2025-26, 11th, Page No. 144)

70. (3)
Parthenium or carrot grass that came into India as a contaminant with imported wheat, has become ubiquitous in occurrence and causes pollen allergy.
(NCERT Ed. 2025-26, 12th, Page No. 217)

71. (1)
Turgidity of cells helps in extension growth.
Water provides the medium for enzymatic activities needed for growth.
(NCERT Ed. 2025-26, 11th, Page No. 171)

72. (3)
In a majority of aquatic plants such as water hyacinth and water lily, the flowers emerge above the level of water and are pollinated by insects or wind as in most of the land plants.
(NCERT Ed. 2025-26, 12th, Page No. 23)

73. (3)
Kinetin does not occur naturally in plants. Search for natural substances with cytokinin-like activities led to the isolation of zeatin from corn-kernels and coconut milk.
(NCERT Ed. 2025-26, 11th, Page No. 176)

74. (2)
In pollen grains the hard outer layer called the exine is made up of sporopollenin.
(NCERT Ed. 2025-26, 12th, Page No. 18)

75. (3)
An increase in surface area denotes the growth in a dorsiventral leaf.
(NCERT Ed. 2025-26, 11th, Page No. 168)

[Contd...

76. (3)

The thymus is degenerated in old individuals, resulting in decreased thymosin production. Since thymosins differentiate T-lymphocytes (cell-mediated immunity), their decrease weakens the immune response.

(NCERT Ed. 2025-26, 11th, Page No. 243)

77. (2)

Catecholamines (Adrenaline/Noradrenaline) increase (not decrease) the heart beat, strength of heart contraction, and rate of respiration. They also stimulate glycogen breakdown to increase blood glucose.

(NCERT Ed. 2025-26, 11th, Page No. 244)

78. (2)

The 11th and 12th pairs of ribs are called floating ribs. The 8th, 9th, and 10th pairs are false ribs.

Ribs are bicephalic because their dorsal end has two articulating surfaces: one with the vertebral body and other with the transverse process of the vertebra.

(NCERT Ed. 2025-26, 11th, Page No. 225)

79. (1)

The seminiferous tubules of the testes open into the vasa efferentia through the rete testis. The vasa efferentia then leave the testis and open into the epididymis located along the posterior surface of each testis. The epididymis leads to the vas deferens that ascends to the abdomen and loops over the urinary bladder.

(NCERT Ed. 2025-26, 12th, Page No. 27)

80. (1)

List-I	List-II
Hypothalamus	Urge for eating and drinking
Cerebellum	Convoluted surface for more neurons
Corpus callosum	Connects cerebral hemispheres
Medulla	Controls respiration and gastric secretions

(NCERT Ed. 2025-26, 11th, Page No. 236)

81. (3)

In Class Aves (birds), the endoskeleton is fully ossified (bony) and the long bones are hollow with air cavities (pneumatic).

(NCERT Ed. 2025-26, 11th, Page No. 49)

82. (1)

Myasthenia gravis is an autoimmune disorder where antibodies attack acetylcholine receptors at the neuromuscular junction.

Muscular dystrophy is a genetic disorder.

Osteoporosis is a degenerative/metabolic bone disease.

Gout is a metabolic disorder caused by uric acid. crystal deposition, not autoimmune.

(NCERT Ed. 2025-26, 11th, Page No. 227)

83. (3)

Coelenterates (Cnidaria) → radial symmetry

Ctenophores → radial symmetry

Echinoderms → radial symmetry in adults (larvae are bilateral)

Annelids, Arthropods, Molluscs, and Chordates show bilateral symmetry

Platyhelminthes are also bilaterally symmetrical

Sponges are mostly asymmetrical

(NCERT Ed. 2025-26, 11th, Page No. 38)

84. (3)

The joint between the atlas (C1) and axis (C2) is a pivot joint.

It allows rotation of the head, enabling movements like shaking the head no. This rotation occurs around the odontoid process (dens) of the axis.

(NCERT Ed. 2025-26, 11th, Page No. 227)

85. (1)

In humans, the pars intermedia is poorly developed (rudimentary) and largely fused with the pars distalis.

While the pars intermedia does secrete MSH (melanocyte-stimulating hormone), MSH is not a trophic hormone.

Trophic hormones act on other endocrine glands (e.g., TSH, ACTH, FSH, LH), whereas MSH acts on melanocytes in the skin.

(NCERT Ed. 2025-26, 11th, Page No. 241)

86. (2)

hCG, hPL and relaxin are produced in women only during pregnancy

Relaxin is secreted by the ovary during the later phase of pregnancy.

During pregnancy, levels of estrogens and progestogens increase several folds in maternal blood. These hormones rise sharply to maintain the uterine lining, suppress ovulation, and support pregnancy. They help in placenta maintenance, uterine growth, mammary gland development, and fetal nourishment.

(NCERT Ed. 2025-26, 12th, Page No. 37)

[Contd...

87. (2)

In Cephalochordata (e.g., Branchiostoma / *Amphioxus*), the notochord extends from head to tail and persists throughout life.

Urochordata / Tunicata → notochord is present only in the larval tail, not in adults.

Vertebrata → notochord is present in the embryo but later replaced by the vertebral column.

(NCERT Ed. 2025-26, 11th, Page No. 46)

88. (1)

The female accessory ducts include the fallopian tubes (oviducts), the uterus, and the vagina.

The womb (uterus) is supported by ligaments, its shape is described as an inverted pear.

The endometrium is the innermost glandular layer of the uterus that breaks down and regenerates periodically as part of the menstrual cycle.

The myometrium is the middle thick layer of smooth muscles responsible for the powerful contractions (labor) required for parturition.

(NCERT Ed. 2025-26, 12th, Page No. 30)

89. (2)

Cyclostomes have a sucking/circular mouth without jaws. Their body is devoid of scales and paired fins. A key behavioral trait mentioned is that they are marine but migrate to fresh water for spawning, after which they die.

(NCERT Ed. 2025-26, 11th, Page No. 47)

90. (1)

The electrical potential difference across a resting plasma membrane is called the resting potential (about –70 mV in neurons).

Action potential → potential during impulse transmission

Nerve impulse → the signal itself

Depolarization → loss/reversal of resting potential

(NCERT Ed. 2025-26, 11th, Page No. 233)

91. (4)

List-I	List-II
Sycon	Choanocytes
Pleurobrachia	Comb plates
Pila	Radula
Balanoglossus	Proboscis gland

(NCERT Ed. 2025-26, 11th, Page No. 40, 42, 44, 45)

92. (2)

Neurons are the functional units of the nervous system that detect, receive, and transmit stimuli.

Schwann cells and glial cells provide support, protection, and insulation.

Nissl's granules are cell organelles involved in protein synthesis.

(NCERT Ed. 2025-26, 11th, Page No. 231)

93. (3)

PTH increases Ca^{2+} levels in the blood by stimulating bone resorption and reabsorption by renal tubules. TCT, conversely, regulates (lowers) blood calcium levels.

(NCERT Ed. 2025-26, 11th, Page No. 243)

94. (4)

White muscle fibers (fast-twitch fibers) have low myoglobin content, fewer mitochondria, well-developed sarcoplasmic reticulum that enables rapid Ca^{2+} release for quick, powerful contractions, depend mainly on anaerobic glycolysis.

Red muscle fibers (slow-twitch fibers) have high myoglobin content, many mitochondria, depend on aerobic metabolism.

(NCERT Ed. 2025-26, 11th, Page No. 223)

95. (4)

The endometrium is essential for implantation. The corpus luteum degenerates without fertilization, causing the endometrium to break down. The reason why the endometrium is necessary for implantation is related to its thickness and vascularity (which provides support), not the fact that it disintegrates when fertilization fails.

(NCERT Ed. 2025-26, 12th, Page No. 35)

96. (3)

Primary oocyte (present from birth till puberty) are diploid (2n), arrested in prophase I

Secondary oocyte (formed after completion of meiosis I) are haploid (n).

First polar body (formed along with secondary oocyte) are haploid (n)

Ovum (formed after completion of meiosis II) are haploid (n).

Second polar body (formed after meiosis II) are haploid (n)

(NCERT Ed. 2025-26, 12th, Page No. 33)

97. (2)

Prolactin regulates the growth of mammary glands and formation of milk, Oxytocin stimulates the vigorous contraction of the uterus and milk ejection from the mammary gland.

(NCERT Ed. 2025-26, 11th, Page No. 242)

[Contd...

98. (2)

Platyhelminthes are triploblastic (have ectoderm, mesoderm, and endoderm) but acoelomate (no body cavity).

Aschelminthes (Nematoda) are triploblastic but pseudocoelomate.

Annelids are triploblastic and coelomate.

Poriferans are diploblastic (or lack true tissue organization), not triploblastic.

(NCERT Ed. 2025-26, 11th, Page No. 42)

99. (1)

List-I	List-II
Leydig Cells	Synthesise and secrete androgens
Sertoli Cells	Provide nutrition to germ cells
Primary Spermatocytes	Periodically undergo meiosis
Secondary Spermatocytes	Formed after the first meiotic division

(NCERT Ed. 2025-26, 12th, Page No. 27, 31)

100. (2)

Acromegaly occurs due to over-secretion of growth hormone (GH) in adults, after epiphyseal closure. It leads to severe disfigurement, especially of the face, hands, and feet (enlarged jaw, nose, lips, and thickened facial features).

Gigantism occurs in childhood and causes excessive height rather than facial disfigurement.

Dwarfism results from low GH and causes short stature, not disfigurement.

Goitre causes neck swelling due to thyroid enlargement, not facial disfigurement.

(NCERT Ed. 2025-26, 11th, Page No. 241)

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